

Exercises

1 $\lim_{x \rightarrow 0} \sqrt{x} \log(x)$

$$\lim_{x \rightarrow 0} \frac{\log(x)}{x^{-1/2}}$$

$$\lim_{x \rightarrow 0} -2\sqrt{x} = 0 \quad \checkmark$$

2 $I_2 = \frac{\sqrt{\pi}}{2} \left(\frac{2}{\sqrt{\pi}} \int_0^2 e^{-x^2} dx \right) = \frac{\sqrt{\pi}}{2} \operatorname{erf}(2)$

3 $E_T \leq \int_0^{\infty} \left| \frac{\sin x}{x^2} \right| dx \leq \int_0^{\infty} \frac{1}{x^2} dx$

$$\int_0^{\infty} x^{-2} dx = \left[-\frac{1}{x} \right]_0^{\infty} = 0 - \left(-\frac{1}{0} \right) = \frac{1}{0} \quad E_T \leq \frac{1}{0}$$

4 I_4 : Do not use Monte Carlo because this is a 2D integral over a rectangular square. Simpson rule will be more efficient and faster.

Is: Use Monte Carlo because the integral is over a domain defined by $x_1^2 + 2x_2^2 \leq 1$. Standard quadrature methods require a complex method to map curved boundaries on a domain.

5 a) $y = e^{-x} \quad x = -\log(y) \quad dx = -\frac{1}{y} dy$

$$\int_0^1 (-\log y) y \sin(-2 \log y) \left(-\frac{1}{y} \right) dy$$

$$I_2 = \int_0^1 (\log y) \sin(2 \log y) dy$$

b) $y = t^2 \quad dy = 2t dt$

$$I_8 \int_0^1 \log(t^2) \sin(2 \log t^2) (2t) dt$$